



Viral Gastroenteritis

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Learning Outcomes

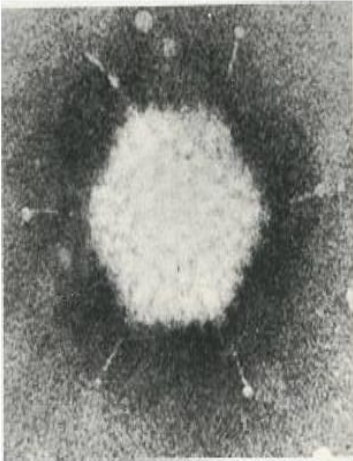
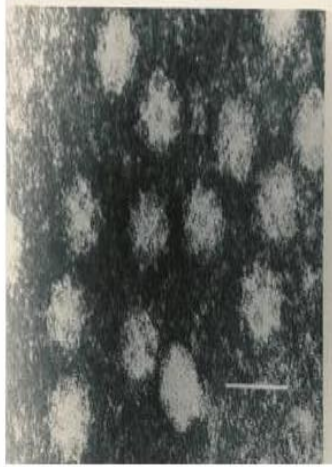
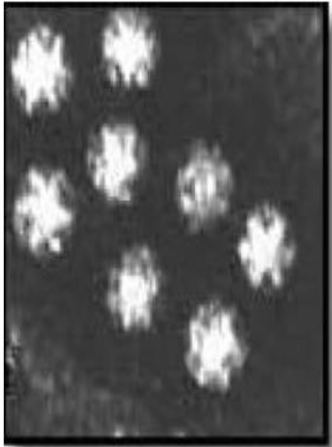
- Identify the most common viruses that can cause infections of the GI tract
- Compare the major characteristics of specific viral diseases affecting the GI tract
- Describe the diagnosis of viral gastroenteritis.

Introduction :

- Acute gastroenteritis is a common cause of morbidity and mortality worldwide.
- Acute viral diarrheal disease is generally self-limiting in industrialized nations but can have significant morbidity for young and older adult patients.
- It is inflammation of the gastrointestinal tract leading to acute diarrhoea and vomiting.
- **Causes:** infectious : (bacteria, viruses, parasites) or non infectious (side effect of some drugs or allergy) .
- **Viral etiology:**
 - Rotavirus
 - Adenovirus 40,41 types
 - Calicivirus
 - Astrovirus

Viral Gastroenteritis

Viral etiology:

Virus	Rotavirus MOST COMMON	Adenovirus 40,41 types 2 nd common	Calicivirus	Astrovirus
Genome	dsRNA	dsDNA	ssRNA (+)	ssRNA (+)
Morphological features	Double shelled with wheel like structure	Classical icosahedral capsid with fibers	Cup like depression on its surface	5 or 6 pointed star on its surface
Pictures:				

The clinician encounters acute viral gastroenteritis in 3 settings:

1-Sporadic gastroenteritis in **infants**, which most frequently is caused by **rotavirus**.

2-Outbreaks of gastroenteritis, which occurs in communities (families, institutions, ships, vacation spots) as a result of classic food-borne or water-borne pathogens . Most of these infections are caused by **Caliciviruses**.

3-Sporadic acute gastroenteritis of **adults**, which most likely is caused by Caliciviruses, Rotaviruses, Astrviruses, or Adenoviruses.

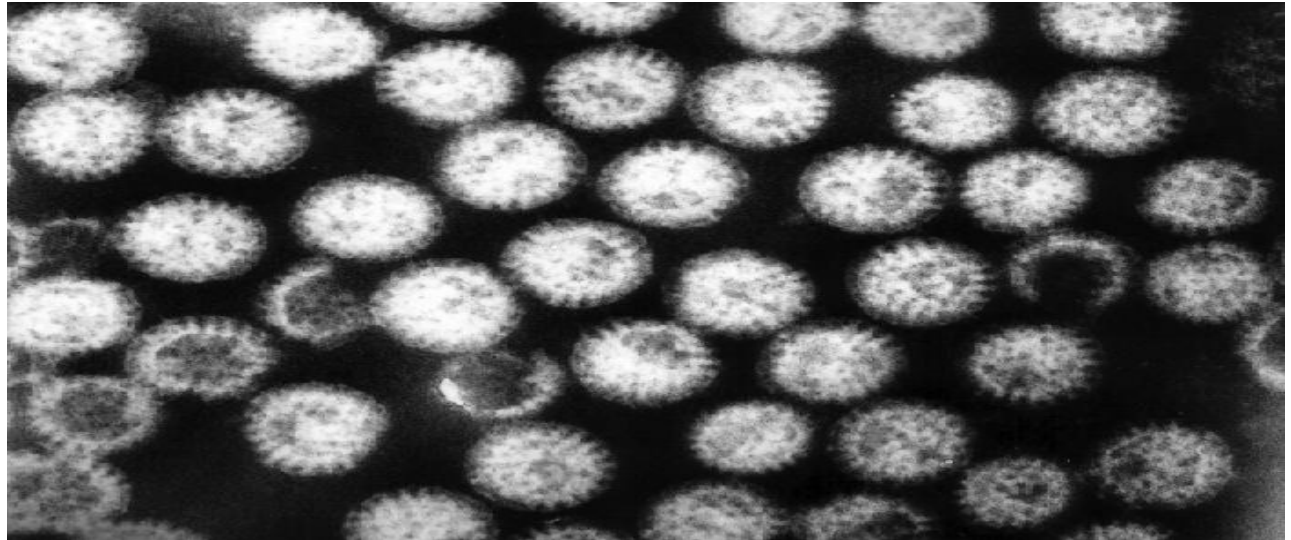
Rotavirus

Rotavirus

Family:	<i>Reoviridae</i> [Respiratory & Enteric Orphan]
Description:	• 11 segments ds-RNA • Double-layered icosahedral capsid • Non-enveloped • ~ 70 nm • RNA – dependent RNA polymerase⁵ • 7 groups [A-G] only A-B-C affect humans- • Group A → most common 95% of the cases
Epidemiology	
Spread (or mode of transmission):	Fecal-oral route through contaminated food or water
Age:	all age groups
Symptomatic infection:	mostly in infant 6 -24 months.
Peak:	Winter months
Infection:	Endemic

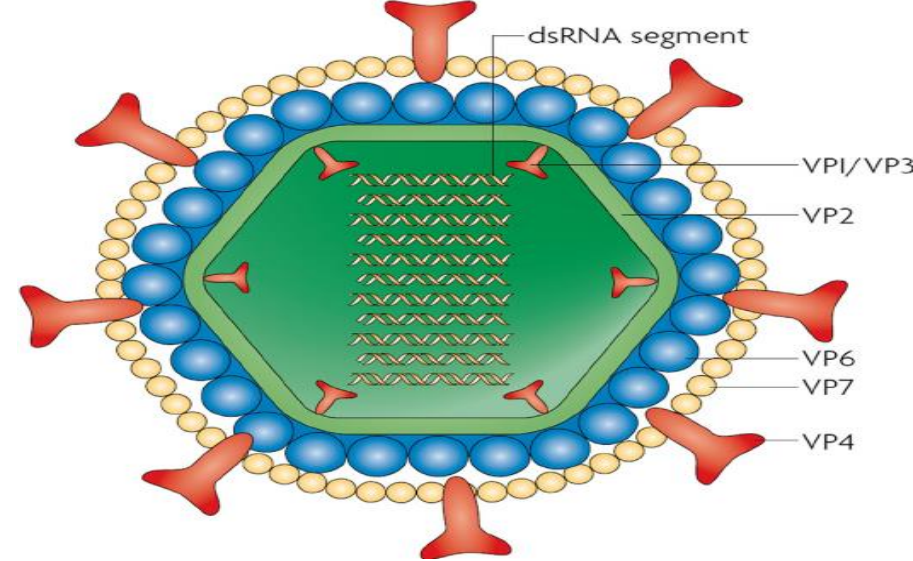
Epidemiology

- Differences in Groups
 - Group A infections are most common
 - Group B has been associated with outbreaks in adults in China
 - Group C is responsible for sporadic cases of diarrhea in infants around the world

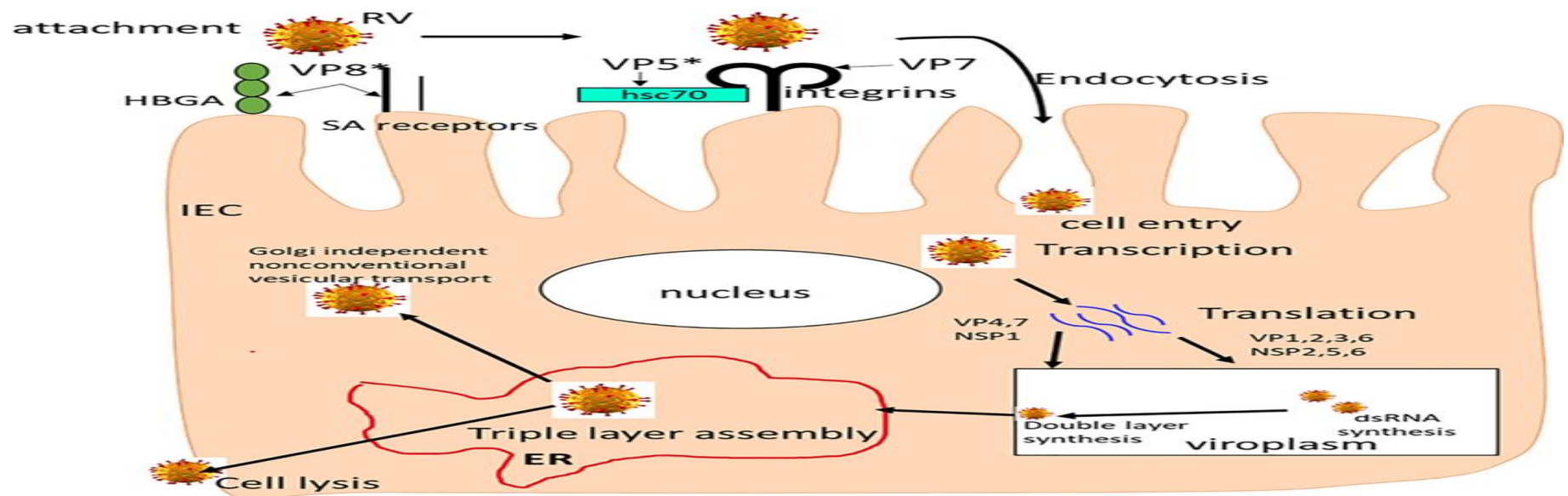


Rotavirus

- Family :Reoviridae
- Virus :Rotavirus
- Characteristics : genome of 11 segments of double stranded RNA.
- The virion has icosahedral symmetry and is triple-layered capsid lacking an envelope



- The outer capsid has two layers, an intermediate layer consisting of the major capsid protein (VP6) and
- An outer layer that contains the viral attachment protein (VP4) and glycoprotein (VP7)
- Outer capsid proteins VP4 and VP7 carry epitopes important in neutralizing activity of Antibody , with VP7 glycoprotein being the predominant antigen.

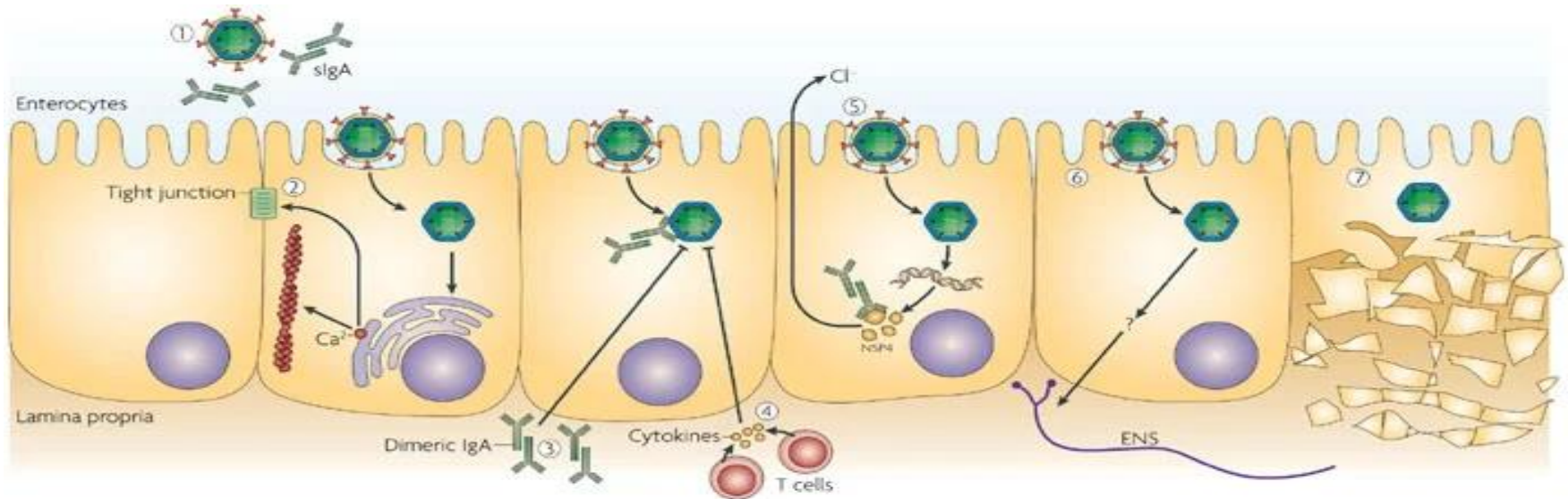


Pathogenesis

- Rotavirus pathogenesis is complicated, with several possible mechanisms, including malabsorption from mucosal damage, viral enterotoxin secretion, and enteric secretions in response to the virus. Rotavirus increases electrolyte secretion from the small intestine and decreases the glucose cotransport of these electrolytes.

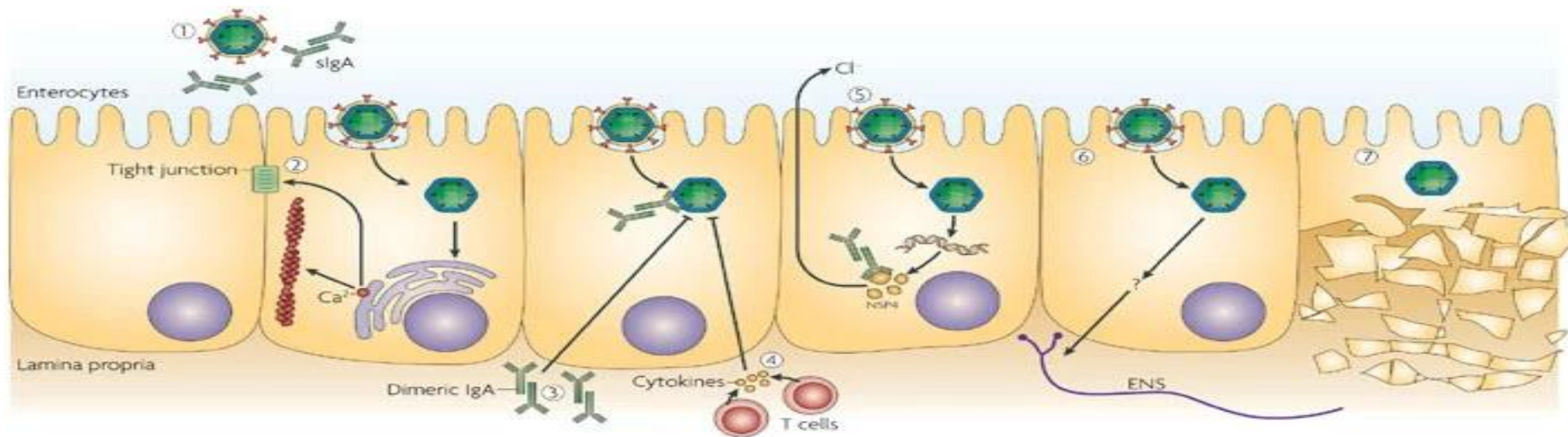
Viral Replication

- Rotaviruses infect cells in the villi of the small intestine.
- They multiply in the cytoplasm of enterocytes and damage their transport mechanisms.
- Rotavirus infection prevents the absorption of water, causing a net secretion of water and loss of ions, which together result in watery diarrhea.



Viral Replication

- The NSP4 protein of rotavirus acts in a toxin-like manner to promote calcium ion influx into enterocytes, the release of neuronal activators, and a neuronal alteration in water absorption.
- The loss of fluids and electrolytes can lead to severe dehydration and even death if therapy does not include electrolyte replacement.



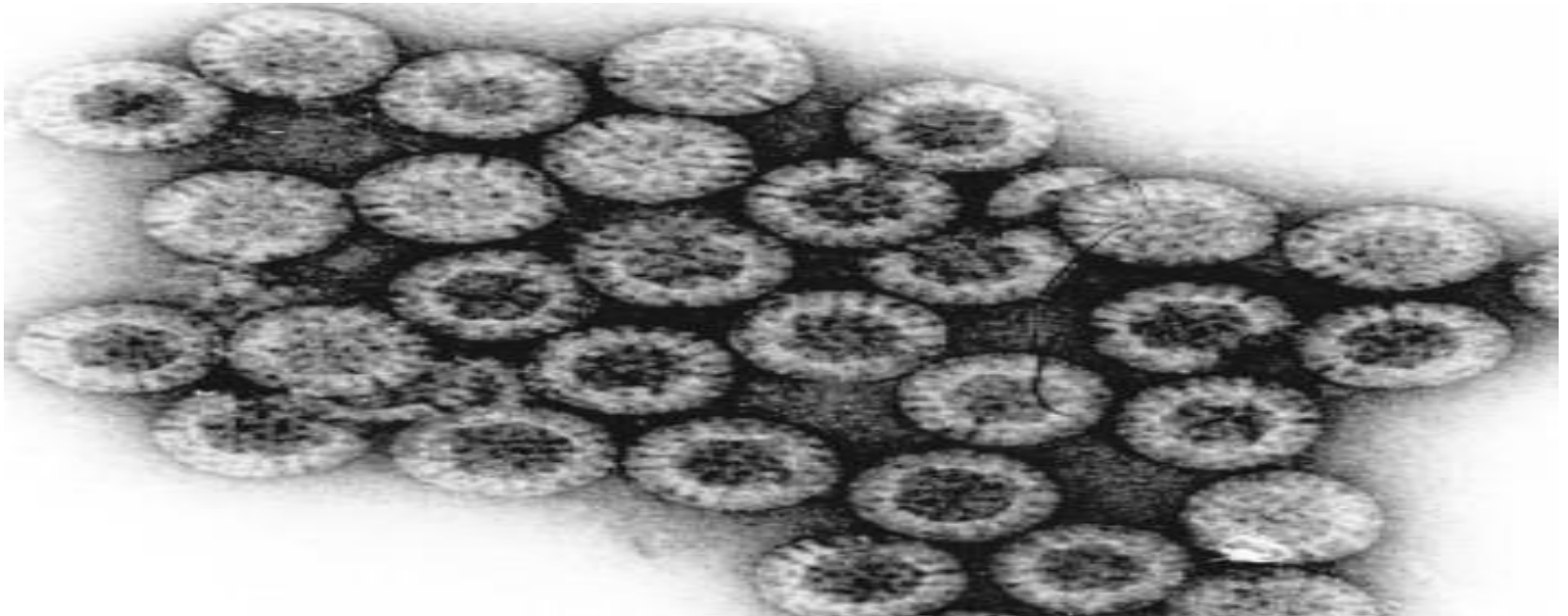
Immunity

- Immunity to infection requires the presence of antibody, primarily immunoglobulin A (IgA), in the lumen of the gut.
- Antibodies to the VP7 and VP4 neutralize the virus.
- Rotavirus diarrhea is self-limiting and patients recover completely within 5-10 days.
- It causes gastroenteritis in adults Immunity to rotavirus infection is unclear. It is likely that intestinal IgA directed
- specific serotypes protects against reinfection and that colostrum IgA
- protects newborns up to the age of 6 months.

Transmission: The primary mode of transmission is predominantly fecal-oral.

Ingestion of contaminated water or food, or contact with contaminated surfaces. Outbreaks can more easily occur among children in day care settings.

Rotaviruses are stable to heat at 50°C, to a 3.0–9.0 range of pH, and to lipid solvents, such as ether and chloroform, but they are inactivated by 95% ethanol, phenol, and chlorine.



Treatment

- The best way to protect a child from rotavirus is with a rotavirus vaccine.
- Hospitalized patients with the disease must be isolated to limit the spread of the infection to other susceptible patients.
- Good hygiene like hand washing and cleanliness are important to control the spread of the disease.

Vaccine :Rota Shield (withdrawn)

RotaTeq(RV5)

Rotarix(RV1)

- Clinical Findings :

The incubation period is about 1–3 days. It has an abrupt onset, characterized by vomiting followed diarrhea, fever and abdominal pain .

- Laboratory diagnosis :

Specimen

Test

of results

Feces

RT-PCR

If positive indicates current infection. is the most sensit

method for

detection of rotavirus from st

ELISA

If positive indicates current

Dip stick/rapid test

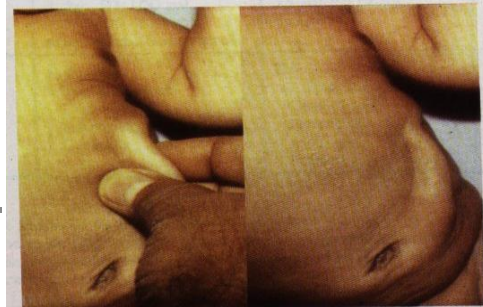
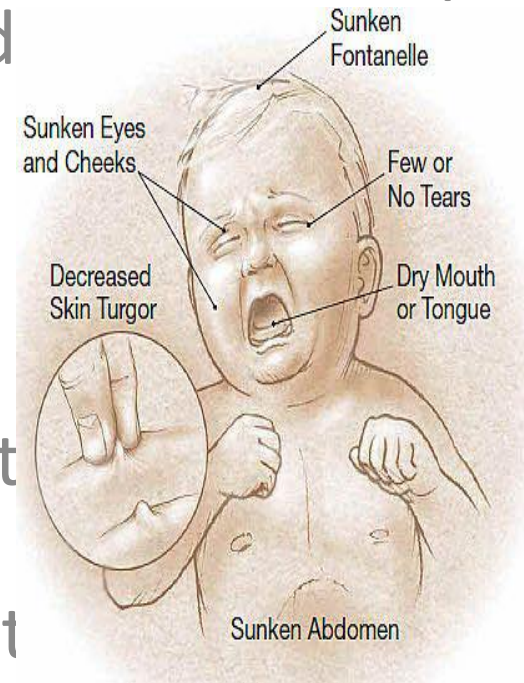
If positive, indica

Electron microscopy

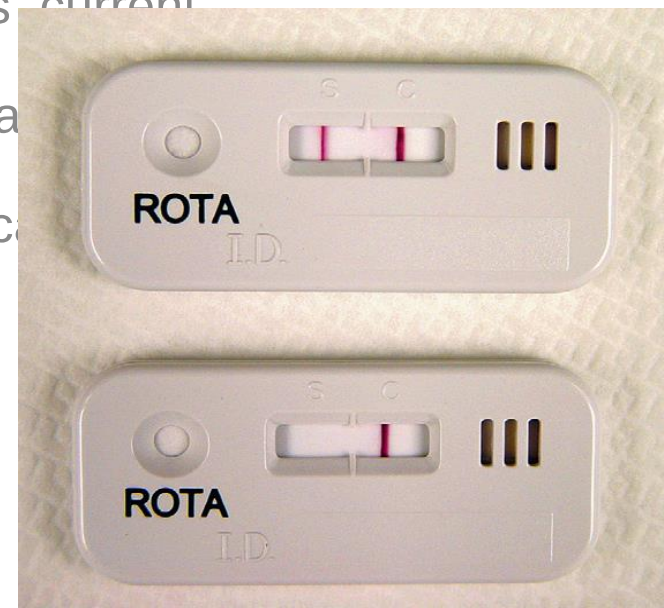
If positive, indic

infection.

- Live attenuated virus vaccine .

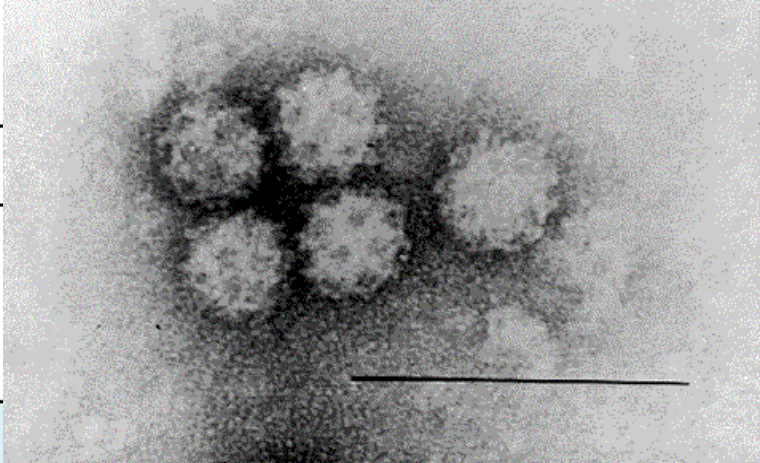


Skin turgor in severe dehydration.



• Caliciviruses:

Family:	Caliciviridae [Calyx =cup]
Description:	• Non-enveloped • ss-RNA with +ve polarity • Icosahedral capsid



Two morphologic types:	
Typical caliciviruses	Small Rounded smooth Structured Viruses
• Sapoviruses	Noroviruses (most common cause of non-bacterial outbreaks of GE worldwide)

Norovirus (Norwalk virus):

Epidemiology:	• Fecal-oral route [water, shellfish(major source)] • Outbreaks of GE in schools, camps & cruises • All age groups
Clinical features:	• Children: vomiting [projectile] • Adults: diarrhea
Diagnosis:	• Viral Ag in stool samples by ELISA



HUMAN CALICIVIRUSES : (HuCV)

- **Family Caliciviridae** : comprises of four genera; out of which two consists of important agents of human Diarrhea

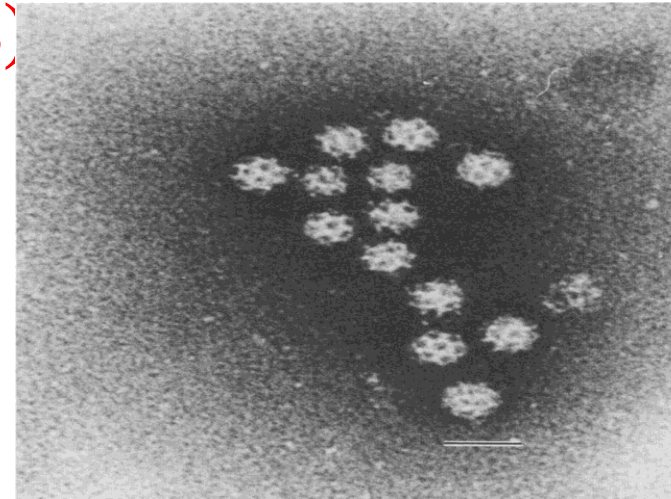
(1)Genus *Norovirus*, which includes the Norwalk viruses, and (2) genus *Sapovirus*.

- Norwalk virus is the prototypic member of the Genus *Norovirus*.
- The cause of epidemic acute gastroenteritis. It was first detected in stools of patients with gastroenteritis in Norwalk, Ohio in 1972.

Caliciviruses

Non-enveloped, ss RNA viruses . *cup-shaped depressions on surface* .(calici= i.e. cup-like).

- cup-shaped depressions on their surfaces.
- single-stranded, positive-sense RNA
- The virion has no envelope.
- Important human pathogens are the noroviruses (eg, Norwalk virus)



Norovirus

- member of the calicivirus family
- Norovirus is a single-stranded RNA member of the calicivirus family.
- This viral pathogen is the most common cause of epidemic diarrheal illness, accounting for >90% of viral gastroenteritis outbreaks and approximately 50% of cases worldwide

- It is common in winter months in temperate climates; therefore called as **winter vomiting disease** or gastric flu
- Low infective dose (~10 virions).
- There is no vaccine
- Viral excretion in stool can continue beyond the cessation of symptoms.
- Outbreaks in institutions, cruise ships, etc. through contaminated food or water (feco-oral) .

Common food sources include contaminated salad, fresh fruits, shellfish (such as oysters), or water. Other sources include the infected person; touching contaminated surfaces.

- There are many animal Caliciviruses, but there is no evidence that they
- cause human infection.

Resistance to inactivation by chlorination and to drying in the environment. It is thought to remain infectious for several days in water, uncooked food, and on environmental surfaces such as door handles.

Diagnosis- Human Calicivirus

- Specimen- stool , vomitus, environmental swabs.
- Immune EM
- RT-PCR in state public health labs.
- Serology for epidemiologic purposes

Treatment and Management

- There is no specific antiviral treatment available for Calicivirus infections.
- Usually self-limited, resolving on its own.
- Management focuses on oral fluid and electrolyte replacement to counteract fluid loss from diarrhea and vomiting.

Enteric Adenoviruses :

Adenoviruses (types 40 and 41) are the **second** most common viral agents of endemic diarrheal illness of infants and young children worldwide <4 years of age .

- Fecal-oral spread .

- Incubation period is 8-10 days .intussusception, mesenteric adenitis, appendicitis .

- Diarrhea lasts 5-12 days, longer than other viral gastroenteritis.

- Diagnosis: Latex agglutination, ELISA, EM.

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Respiratory viruses:

Diarrhea has also been reported as a part of manifestations of certain respiratory viruses such as:

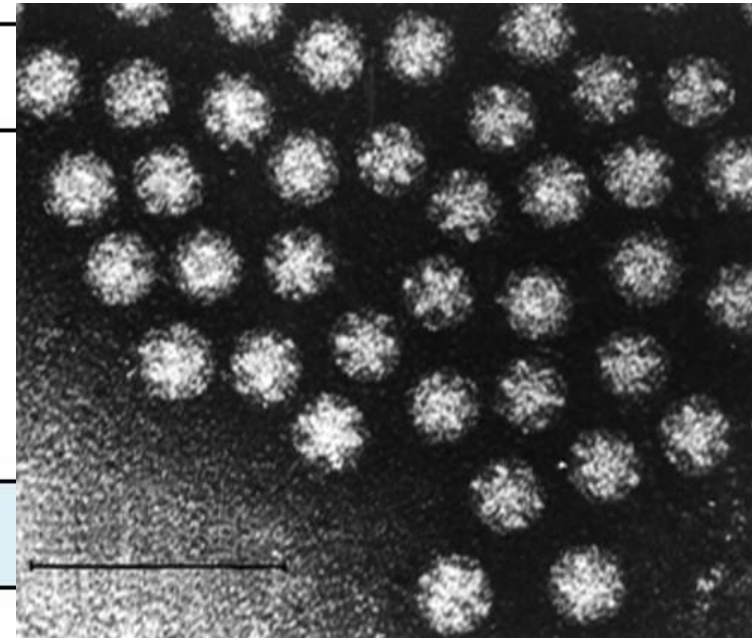
SARS coronaviruses

Influenza A/H5N1 virus

Influenza A/H1N1 virus (the 2009 pandemic strain).

Astrovirus (*Astroviridae*)

Family:	Astroviridae [astro= a star]
Description:	• Non-enveloped • ss-RNA with +ve polarity • Icosahedral capsid • 8 serotypes



Clinical feature:

- Mild GE
- Outbreak of diarrhea <5 ys. (usually endemic)

Lab Diagnosis:

Ag detection in stool by ELISA

Astrovirus - Epidemiology

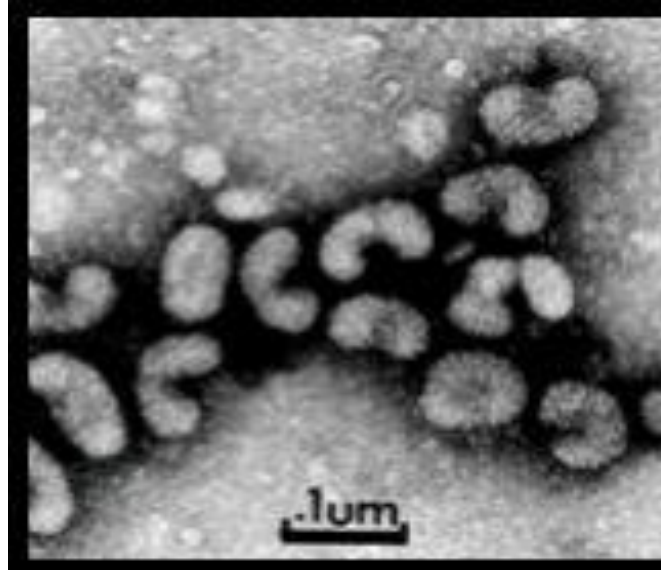
- Worldwide
- Mainly in children <7 years of age.
- Transmission person-to-person via
- Fecal-oral route of transmission, contaminated oysters
- Outbreaks due to fecal contamination of sea-food or water

Astrovirus – Diagnosis

- RT-PCR is the **gold-standard** method .
- EM (virus shed in stool in great numbers)
- EIA/ELISA EIA/ELISA (Enzyme Immunoassay/Enzyme-Linked Immunosorbent Assay)
- Cell Culture:

TOROVIRUS

- ◆ Family Coronaviridae
- ◆ Genus Torovirus
- ◆ Human and animal pathogen
- ◆ Coated ss (+) RNA virus
- ◆ Core “doughnut-shaped” (torus)
- ◆ Watery diarrhea in 2 –12 months old
- ◆ Diagnosis: EM



Thank you